

Scala For DS

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Class : SYIT	Assignment Date : 18/08/2026

Practical no.19

19) Combine two plots (e.g., scatter and line plot) in a single visualization using Breeze-viz.

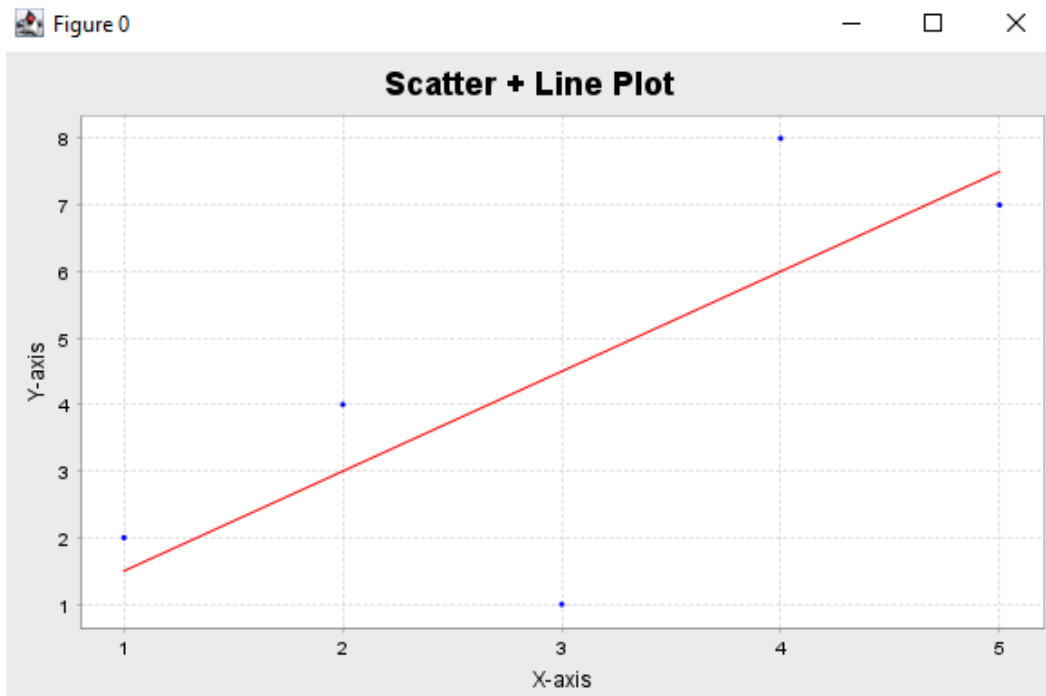
INPUT:

```
name := "PRACTICAL"
version := "0.1"
scalaVersion := "2.12.20" // Change to match your Scala version
lazy val root = (project in file("."))
  .settings(
    name := "d21",
    libraryDependencies += Seq(
      "org.scalanlp" %% "breeze" % "2.1.0",
      "org.scalanlp" %% "breeze-viz" % "2.1.0"
    )
  )
```

```
import breeze.linalg._
import breeze.plot._

object PRACTICAL extends App {
  val x = DenseVector(1.0, 2.0, 3.0, 4.0, 5.0) // Sample X-axis values
  val yscat = DenseVector(2.0, 4.0, 1.0, 8.0, 7.0) // Y values for scatter
  val yline = DenseVector(1.5, 3.0, 4.5, 6.0, 7.5) // Y values for line plot
  val fig = Figure() // Create a new figure
  val plt = fig.subplot(0)
  plt += plot(x, yscat, '.', colorcode = "blue")
  plt += plot(x, yline, '-', colorcode = "red")
  plt.title = "Scatter + Line Plot"
  plt.xlabel = "X-axis"
  plt.ylabel = "Y-axis"
  fig.refresh()
  println("S007 DHRUV")
}
```

OUTPUT:



Practical no.20

20) Read a CSV file and calculate basic statistics for each numeric column. Use the scala-csv library or similar tools.

INPUT:

```

import scala.io.Source
object PRACTICAL extends App {

    val file = Source.fromResource("iris.csv")
    val data = file.getLines()
        .drop(1)
        .map(line => line.split(",")(0).toDouble)
        .toList

    file.close()
    val count = data.length
    val mean = data.sum / count
    val min = data.min
    val max = data.max
    val variance = data.map(x => math.pow(x - mean, 2)).sum / count
    val std = math.sqrt(variance)

    println(s"Count: $count")
    println(f"Mean: $mean%.2f")
    println(f"Minimum: $min%.2f")
    println(f"Maximum: $max%.2f")
    println(f"Standard Deviation: $std%.2f")
    println("S007 DHRUV")
}

```

OUTPUT:

```

Run PRACTICAL x
C:\Program Files\Eclipse Adoptium\jdk-21
Count: 150
Mean: 75.50
Minimum: 1.00
Maximum: 150.00
Standard Deviation: 43.30
S007 DHRUV
Process finished with exit code 0

```